VARUN R YATTINAHALLI

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Education

Dayananda Sagar Academy of Technology and Management — **Expected 2027**
B.E. Computer Science and Engineering (Data Science) — CGPA: 9.0 — Bengaluru, India

Vishwachetana Vidyanikethana PU College — **2022 – 2023**
Higher Secondary Education — 95.83%

Kourava Education Society — **2020 – 2021**
Secondary Education — 95.04%

Projects

Enterprise Semantic Data Classification System | Python, SBERT, NLP, LLM Integration — **Apr 2026 – May 2026**

- Engineered an AI-driven semantic classification system for enterprise datasets across Banking, Insurance, Healthcare, and Petroleum domains.
- Applied NLP techniques and semantic similarity methods using SBERT to automate schema understanding and data categorization across enterprise datasets.
- Implemented lexical analysis, semantic matching, and domain-specific mappings to enhance classification robustness across Banking, Insurance, Healthcare, and Petroleum domains.
- Integrated a hybrid SBERT-LLM framework that improved classification accuracy from 47% to 97% while eliminating unknown classifications and timeout errors on a 200-case evaluation dataset.

Renewable Energy Utilization Prediction | Python, Machine Learning, Pandas, NumPy — **Mar 2025 – May 2025**

- Designed a time-series forecasting solution for renewable energy resource utilization using historical energy datasets.
- Performed data cleaning, feature engineering, and exploratory data analysis to prepare data for predictive modeling.
- Built and evaluated a Polynomial Regression (Degree 2) model that achieved 96% predictive performance on the target dataset.
- Analyzed forecasting metrics and identified utilization patterns to support data-driven decision making.

Real-Time Blind Assistant | Python, YOLOv8, Ollama Phi-3, Computer Vision — **Aug 2025 – Nov 2025**

- Developed a real-time assistive system for visually impaired users using YOLOv8-based object detection and AI-generated voice feedback.
- Integrated Ollama Phi-3 with a mobile camera feed to generate natural language responses from live visual input.
- Implemented real-time detection and audio response workflows to improve accessibility and situational awareness.
- Optimized the application pipeline to achieve smoother interaction and lower inference latency.

Technical Skills

Languages: Python, C++, SQL
Data Science & Analytics: Machine Learning, Natural Language Processing, Data Analysis, Exploratory Data Analysis (EDA), Data Visualization
Concepts: Data Structures and Algorithms, Database Management Systems
Libraries & Tools: NumPy, Pandas, Scikit-Learn, Git, GitHub, Jupyter Notebook

Coding Profiles

- LeetCode – View Profile
- Coding Ninjas – View Profile

Publication

- Co-author of the review paper titled *A Review of Past Discovery, Current Trends, and Future Directions in Brain Computer Interface*, published in Indiana Journal of Multidisciplinary Research, Vol. 4 Issue 3 (2024).